# Hamiltonian Vision Kernel: Symmetry-Pooled Quantum Correlator Features with Hardware Validation

Sparsho Chakraborty* and Siddhartha Patra†

***Abstract*—We introduce the Hamiltonian Vision Kernel (HVK), a new hybrid quantum–classical architecture for image reconstruction. HVK assembles matrix-product-state patch features, a variational quantum circuit exposing a Pauli-correlator latent space, two-dimensional Fourier positional encoding, a learnable Hamiltonian energy diagnostic, and a classical decoder into a single pipeline; HVK1D and HVK2D instantiate chain and grid interaction graphs, and observable pooling gives the HVK2D map $D_4$ equivariance by construction. This specific combination is new, not any one ingredient in isolation (Section II). Across six real image datasets, per-image-trained HVK2D models reconstruct with high fidelity ($25.8$–$41.6$ dB PSNR). On a resource-matched, multi-seed held-out CIFAR-10 comparison, the tested HVK2D map does not exceed simple local-observable or raw-linear classical controls ($18.12$ vs. $18.80$ dB); a pre-declared Two One-Sided Tests (TOST) equivalence test at a $\pm 1$ dB margin confirms this gap is small enough to call the two *competitive* rather than merely indistinguishable, while HVK2D substantially outperforms a random-VQC and a strict-classical-random-feature control. HVK also executes on real IBM Quantum hardware: five fitted checkpoints are replayed directly from hardware-measured observables, with no retraining, reaching $25.90$–$31.52$ dB PSNR. If HVK ties resource-matched classical baselines on ordinary reconstruction, its case rests elsewhere: an entangling observable channel that is representationally necessary on a leakage-audited task requiring distant-patch products (mean $R^2 = 0.974$ vs. $R^2 \leq 0.02$ for non-entangling controls), an interpretable Hamiltonian energy diagnostic, and native execution on quantum hardware – capabilities a purely classical reconstruction pipeline does not provide. We make no claim of quantum advantage: HVK is presented as a new, working architecture that is competitive with resource-matched classical baselines, validated on real hardware, and equipped with capabilities a classical reconstruction pipeline does not natively expose. A companion supplementary study provides the resource-matched held-out controls, leakage audits, statistical tests, and complete hardware methodology behind these claims.**

***Index Terms*—Quantum Image Processing, Tensor Networks, Variational Quantum Algorithms, Pauli Observables, Hamiltonian Diagnostics, Group-Equivariant Representations, Hybrid Quantum Hardware Validation.**

## I. Introduction

Hybrid quantum–classical models for vision tasks are increasingly proposed as a route to representations with favorable inductive bias, exploiting entanglement, physically motivated regularizers, or tensor-network structure that classical convolutional pipelines do not natively expose [1], [2], [20]. The variational quantum eigensolver and its descendants exploit the local decomposability of physically relevant Hamiltonians to construct shallow, problem-aware circuits [1]–[3], and quantum image processing provides encodings that place pixel information directly into qubit registers, from flexible amplitude-style representations [36] to basis-encoded pixel values [37] and their processing primitives [4]–[6]. Tensor-network methods, in parallel, have proven effective as compressed representations of structured classical data, including images [7]–[11], [22], and more broadly other structured data such as particle-physics events [16]. Related hybrid quantum architectures target adjacent imaging tasks, including quantum annealing-based tomographic reconstruction [12], adiabatic quantum computing for emission tomography [14], quantum neural compressive sensing for ghost imaging [13], hybrid autoencoder-plus-quantum-SVM feature extraction for image classification [17], quantum convolutional networks and their classical-data variants [38], [39], and quantum capsule networks [18]. HVK focuses on the specific combination of a Pauli-correlator latent space, enforced group-equivariant pooling, and a physically motivated energy diagnostic.

We introduce the Hamiltonian Vision Kernel (HVK1D/HVK2D), which combines these ingredients into a single architecture with three properties studied jointly here: a *Pauli-correlator latent space* (measured VQC observables, not a raw quantum state, form the representation handed to a classical decoder, in the spirit of quantum-enhanced feature maps [21]); an *exact, numerically-verified $D_4$-equivariant observable-pooling construction*, group-averaging the observable grid over the eight symmetries of the square rather than merely motivating symmetry through architecture choices, in the spirit of group-equivariant classical vision [23], [40] and the equivariant-quantum-neural-network line that formalizes symmetry-embedding constructions for variational circuits [24], [41]–[43]; and a *learnable Hamiltonian energy diagnostic* (full nearest-neighbor $XX/YY/ZZ$ sectors in HVK1D and grid-edge $ZZ$ terms in HVK2D). Table VIII contrasts these properties against classical, adversarial, and standard quantum autoencoder baselines.

*a) Novelty relative to the nearest hybrid-vision architectures.:* HVK's closest architectural relatives are three hybrid quantum-vision patterns already established individually in the literature, and situating HVK against each clarifies what is new here versus what is inherited. *Register-based quantum autoencoders* [28]–[30] encode an image's pixel amplitudes directly into a qubit register and train a trash-qubit fidelity

*School of Basic Sciences, Department of Physics, Indian Institute of Technology Bhubaneswar, Odisha, India.
†Assistant Professor, Centre for Quantum Engineering, Research and Education (CQuERE), TCG CREST, Kolkata, India.

objective (or attach a classical decoder to that register's measured amplitudes); HVK never amplitude-encodes pixels into qubits at all — it exposes only classically pre-compressed MPS patch features as circuit input, and hands the decoder measured Pauli correlators rather than register amplitudes or a fidelity score. *Quantum-kernel vision models* [21], [31] embed an input into a quantum feature space and read out a scalar kernel (state-overlap) value consumed by a classical kernel method; HVK instead reads out an explicit vector of local and pairwise Pauli expectation values as a latent representation for a trained decoder, and adds enforced group-equivariant pooling and a Hamiltonian regularizer that a kernel construction has no analogue for. *Tensor-network (TN) circuit classifiers* [9], [11], [22] use MPS structure either as a standalone classical model or as a template for a shallow quantum circuit trained for classification; HVK uses MPS compression only as a classical *preprocessing* step that sets a separately trained VQC's rotation angles, and targets per-image reconstruction rather than classification. None of these three families combines classical MPS-compressed input, a measured Pauli-correlator (not kernel- or register-state) latent space, enforced $D_4$-equivariant pooling, and a learnable Hamiltonian diagnostic in one pipeline; HVK's contribution is this specific *assembly* and the properties it jointly enables (Table VIII), not any single ingredient in isolation — each of which individually recurs throughout this literature (Section V-C).

*b) Contributions.:* **1) Architecture:** HVK1D/HVK2D combines MPS patch features, a six-qubit VQC, a 19–27-dimensional Pauli-correlator vector, 2D Fourier positional encoding, a graph-Hamiltonian diagnostic, and a classical decoder.

**2) Reconstruction scope:** Identically configured per-image fitting experiments cover six real image datasets (Section IV-A).

**3) Competitive with resource-matched classical baselines:** A pre-declared TOST equivalence test on held-out CIFAR-10 shows HVK2D is statistically equivalent, within a $\pm 1$ dB margin, to six of eight resource-matched classical/ablated controls, while still beating the two weakest controls by a wide margin (Section IV-C).

**4) Hardware pilot:** Five fitted checkpoints are replayed on IBM hardware without retraining and decoded from hardware-measured observables (Section IV-D), strengthened by a noise/shot-budget sweep and repeated execution across backends (Sections IV-E–IV-F).

**5) Equivariant observable pooling:** An exact, numerically-verified $D_4$-equivariant observable-pooling construction makes the HVK2D observable map equivariant to the eight symmetries of the square, to floating-point precision (Section IV-G). The guarantee follows from Reynolds group averaging over the $D_4$ orbit, not from anything quantum: the identical construction applied to a purely classical feature map yields the same exactness (Supplementary Material). We therefore report it as a property the assembled pipeline provides, not as a quantum result.

**6) Restricted nonlocal result:** In a leakage-audited fixed-readout protocol, entangling pair observables are necessary when the target depends explicitly on distant-patch products (Section IV-H).

A companion document, *Supplementary Material for the Hamiltonian Vision Kernel: Resource-Matched Ablations, Validation, and Reproducibility*, reports resource-matched held-out comparisons of HVK and classical/ablated feature maps, together with full hardware-pilot methodology (circuit construction, validation, job identifiers, quota accounting) and explicit provenance audits of excluded exploratory results, among them the training-dynamics traces of Section IV-I, which we report as an interpretability readout rather than as evidence of a transition. We keep that analysis in a companion document so the architecture-level contributions above and the generalization question do not obscure each other; Section IV-C summarizes its central finding and how it relates to the results here.

## II. Hamiltonian Vision Kernel: Architecture

HVK treats each spatial patch as a finite spin state, compresses it as a matrix product state (MPS), measures a fixed set of Pauli observables from a variational quantum circuit (VQC), and decodes those observables back into pixels with a classical multi-layer perceptron (MLP). A learnable graph-Hamiltonian energy term regularizes training; unlike parameterized Hamiltonian learning approaches that treat the Hamiltonian as the primary learned object [19], HVK's energy term is a diagnostic regularizer alongside a separate reconstruction objective. The shared pipeline is:

$$\begin{aligned} \text{Image} &\rightarrow \{\text{Patches}\} \rightarrow \text{MPS} \\ &\rightarrow \text{46-D/22-D Feature} \rightarrow \text{6-D Vector} \\ &\rightarrow \text{27-D/19-D Latent} \rightarrow \text{Decoder} \\ &\rightarrow \text{Reconstruction.} \end{aligned} \tag{1}$$

Three of these design choices warrant explicit motivation rather than being left implicit in the pipeline above. *Why MPS features*: treated as an amplitude-encoded quantum state, a patch of $n$ sites has a coefficient tensor with $2^n$ entries; matrix-product-state factorization instead represents it as a chain of small tensors bounded by a fixed bond dimension $\chi$, a compact, singular-value-controlled compression whose entanglement-entropy profile is already known to track image compressibility [8], [22], and whose local expectations and correlations are cheap classical features rather than requiring a full state-preparation circuit. *Why a Pauli-correlator latent space rather than a raw quantum state*: handing the classical decoder a fixed vector of measured Pauli expectation values, instead of full state tomography, keeps the quantum-to-classical interface at $O(\text{qubits})$ rather than exponential readout cost and follows the same measured-observable-as-feature pattern used by quantum-enhanced feature maps generally [21]; it is also what makes the interpretability diagnostics of Section V-A and the training-dynamics diagnostics reported in the companion Supplementary Material possible, since a full state vector would not expose a comparably compact, physically meaningful summary. *Why a grid rather than chain interaction topology for HVK2D*: aligning the six-qubit interaction graph with the $2 \times 3$ patch-neighbor adjacency, rather than an arbitrary linear chain, is what makes the $D_4$ group-pooling construction of Section IV-G well-defined in the

TABLE I
HVK VARIANTS TESTED IN THIS PAPER.

| Variant | Topology | Latent dim. | Symmetry |
|---|---|---|---|
| HVK1D | 6-qubit chain | 27-D | None (control) |
| HVK2D | $2 \times 3$ grid | 19-D | $D_4$-motivated only |
| Symmetric HVK1D | 6-qubit chain | 27-D | U(1) coupling |
| $D_4$-pooled HVK2D | $2 \times 3$ grid, pooled | 19-D | $D_4$-equivariant (exact) |

first place — the pooling operator averages over the spatial symmetries of the square, which requires a latent structure that itself respects the patch grid's two-dimensional adjacency. HVK1D's chain topology is retained as the ungridded control this comparison needs (Section V-D), not because a chain is separately motivated for 2D image data.

Two executed preprocessing configurations share this pipeline. For the $256 \times 256$ *Monalisa*/HVK1D benchmark, sixteen non-overlapping $64 \times 64$ patches are $\ell_2$-normalized, reshaped as 12-site spin states, and sequentially SVD-compressed to MPS bond dimension $\chi = 4$. Their 24 local $X/Z$ expectations, 11 nearest-neighbor $ZZ$ correlations, and 11 bipartite entropies form a 46-D feature vector. For the native $32 \times 32$ HVK2D multi-dataset experiments, sixteen $8 \times 8$ patches are represented by six-site MPSs, giving 12 local expectations, five nearest-neighbor $ZZ$ terms, and five entropies (22 dimensions). A learned linear layer compresses either MPS feature vector to six circuit angles. HVK1D then measures 27 observables (six local $Z$, six local $X$, and five each of $ZZ$, $XX$, and $YY$); HVK2D measures 19 (six local $Z$, six local $X$, and seven grid-edge $ZZ$ terms). The respective Fourier positional vectors have dimension eight and four, following the established use of Fourier positional encoding as a spatial inductive bias in classical generative vision models [15]. In both configurations, a two-hidden-layer MLP (128 and 256 units) maps the observable and positional vectors to one grayscale patch. Figure 1 shows the exact HVK1D and HVK2D ansatz circuits (rebuilt in Qiskit and numerically verified against the training-time PennyLane circuit as part of the hardware pilot of Section IV-D); complete Hamiltonian, loss, positional-encoding, and measurement definitions are retained in the companion supplement.

Table I summarizes the tested variants. HVK1D uses a linear 6-qubit chain (5 nearest-neighbor bonds, 27-D observable vector); HVK2D arranges the same 6 qubits on a $2 \times 3$ lattice matching patch-boundary adjacency (19-D latent signature); a symmetric HVK1D variant adds a $\mathrm{U}(1)$-preserving coupling constraint; and a $D_4$-pooled HVK2D variant (Section IV-G) group-averages the observable grid over the eight symmetries of the square, $\Phi_{D_4}(x) = \frac{1}{8}\sum_{g \in D_4} P_g^{-1}\Phi(gx)$, which is equivariant by construction. The IBM Cloud circuit probe reported in the supplement confirms both HVK1D and HVK2D compile to depth-18, 6-qubit circuits on a Heron-class basis.

## III. Experimental Setup

All image-reconstruction results in Sections IV-A–IV-D use the single-image *Monalisa* protocol developed for HVK: a fresh model is optimized directly on its target image, with no held-out split. They therefore measure per-image fitting and reconstruction capability, including hardware replay of those fitted checkpoints. The symmetry and restricted-entanglement diagnostics use the separate protocols stated in their respective sections; the exploratory training-dynamics material is addressed only as a provenance audit and is not retained as evidence. Held-out generalization and component necessity are evaluated under the resource-matched statistical protocol of the companion supplementary study (Section IV-C). All settings report MSE and PSNR $= 20\log_{10}(1/\sqrt{\mathrm{MSE}})$; image settings additionally report SSIM. Full implementation settings and environment versions are given in the companion supplementary document.

TABLE II
HVK2D SAME-SET RECONSTRUCTION ACROSS SIX REAL IMAGE DATASETS (PER-IMAGE TRAINED, 80–100 STEPS, MEAN $\pm$ STD OVER SAMPLED IMAGES).

| Dataset | $n$ images | PSNR (dB) | SSIM |
|---|---|---|---|
| CIFAR-10 | 3 | $37.75 \pm 2.38$ | $0.9986 \pm 0.0003$ |
| MNIST | 3 | $25.77 \pm 2.23$ | $0.9783 \pm 0.0116$ |
| Fashion-MNIST | 3 | $34.55 \pm 3.05$ | $0.9980 \pm 0.0006$ |
| PathMNIST | 3 | $36.86 \pm 2.11$ | $0.9118 \pm 0.0627$ |
| BloodMNIST | 2 | $36.57 \pm 2.36$ | $0.9922 \pm 0.0063$ |
| PneumoniaMNIST | 2 | $41.60 \pm 0.98$ | $0.9985 \pm 0.0006$ |

## IV. Results

### A. Reconstruction Across Six Real Image Datasets

For each image, we trained a fresh HVK2D model using nonoverlapping $8 \times 8$ patches (16 patches/image, 80–100 optimization steps). The sample covers CIFAR-10 [32], MNIST [33], Fashion-MNIST [34], and the PathMNIST, BloodMNIST, and PneumoniaMNIST subsets of MedMNIST v2 [35]. Table II reports mean PSNR between $25.8$ and $41.6$ dB (SSIM $0.91$–$1.00$). This is a same-set fitting result across varied image structures, not a held-out generalization claim; the companion study provides the latter evaluation (Section IV-C).

### B. Scope and Adaptation Across Images

A model optimized on one image alone does not transfer zero-shot to a structurally different image: on *Monalisa*, zero-shot transfer to a second image reaches only $8.31$ dB PSNR (SSIM $= 0.1044$), while extending training to include the second image recovers $28.63$ dB (SSIM $= 0.9858$). This confirms that the per-image results above measure per-image optimization and reconstruction capability, not held-out generalization — consistent with how we scope every claim in this paper, and precisely the distinction the held-out protocol of Section IV-C is designed to test.

### C. Competitive with Resource-Matched Classical Baselines

The results above measure per-image fitting; a separate, equally important question is whether HVK's feature map improves reconstruction on real, *held-out* images under a resource-matched multi-seed protocol. That question is answered in the companion document, *Supplementary Material*

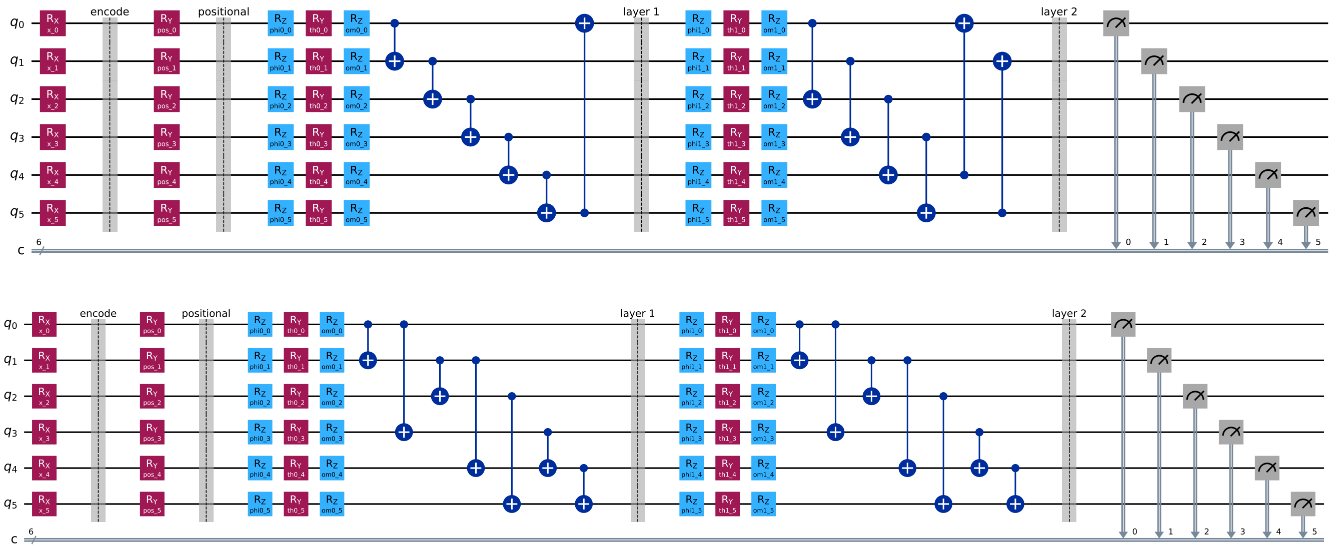


Fig. 1. The HVK1D (top) and HVK2D (bottom) variational ansätze, exactly as executed (encoding $R_X$ rotations, positional $R_Y$ rotations, then two entangling layers of $R_Z$-$R_Y$-$R_Z$ single-qubit rotations with a CNOT ring for HVK1D or fixed grid-edge CNOTs for HVK2D). Both circuits were rebuilt gate-for-gate in Qiskit and validated against the original PennyLane training circuit before use in the hardware pilot of Section IV-D.

*for the Hamiltonian Vision Kernel: Resource-Matched Ablations, Validation, and Reproducibility*. On held-out CIFAR-10, local-observable and raw-linear controls reach $18.80 \pm 1.42$ dB against HVK2D's $18.12 \pm 1.54$ dB; after aggregating held-out images within each split, the HVK2D-minus-control difference is $-0.68$ dB (bootstrap 95% CI $[-0.91, -0.46]$ dB; Wilcoxon $p = 0.0625$, $n = 5$ seeds), with the same numerical direction on five further real datasets. Thus the tested HVK map is informative but is not shown to outperform the simpler maps on ordinary held-out reconstruction. A TOST equivalence test at a declared $\pm 1$ dB margin (companion supplement) confirms this gap is small enough, in a pre-declared and data-independent sense, to call the two *competitive* rather than merely "not shown to differ": six of eight tested controls are statistically equivalent to HVK2D within that margin, while the two weakest controls (strict classical random features, random-VQC) are correctly excluded because HVK2D outperforms them by a wide, practically meaningful margin. We report this as a rigorous complement: it delineates the tested boundary of the positive entanglement and symmetry results reported below and records the provenance checks that prevented exploratory artifacts from becoming claims. A matched-budget, multi-seed rerun of the component-ablation table (240 steps, 3 seeds/variant, replacing an earlier table whose per-seed provenance could not be recovered) reaches the same conclusion on ordinary per-image reconstruction: a plain classical-replacement control ties or slightly exceeds the quantum baseline, while only a circuit-free random-VQC control performs clearly worse; that rerun uses a descriptive pooled-standard-deviation comparison rule at 3 seeds rather than a calibrated test, and full results are in the companion supplement. Section V-B discusses why this ties-with-classical result and the entanglement-necessity result of Section IV-H are not in tension.

### *D. A Real Hardware Reconstruction Pilot*

Every result above is a noiseless-simulator result, a limitation we flagged rather than left implicit. We report a first step past it: a small but complete image-reconstruction pilot executed on real IBM Quantum hardware (`ibm_fez`, a 156-qubit Heron-class processor), beyond the compiled-circuit feasibility probe documented in the supplement. For five images — the full 16-patch *Monalisa* benchmark and four native-resolution CIFAR-10 images (*cat*, *ship*×2, *airplane*), each optimized directly on its own target image following the same per-image protocol as every result in this paper — we took an already-trained HVK1D/HVK2D checkpoint's exact learned parameters, rebuilt its measurement circuits gate-for-gate in Qiskit, verified the translation against the original PennyLane simulator to within numerical noise ($\leq 7.9 \times 10^{-4}$ maximum absolute error per observable across every patch; full validation numbers in the companion supplementary document), and then executed the same circuits on real hardware with 256 shots per measurement basis, a fixed per-basis shot budget rather than an adaptive shadow-tomography-style estimation scheme [25]. The hardware-measured observables were decoded by the same trained classical decoder, with no retraining and no simulator involved anywhere in the reconstruction path.

Reconstructions from real, noisy hardware measurements retain recognizable image structure. *Monalisa* reaches 25.90 dB PSNR from hardware measurements, against a 37.99 dB noiseless-simulator replay of the identical circuit (Figure 2); the four CIFAR-10 images average $29.56 \pm 2.03$ dB on hardware against $42.69 \pm 3.84$ dB in simulation (Figure 3, Table III). The observed gap is consistent with noise in these unmitigated device executions, although the five-image pilot does not isolate individual noise sources. Every hardware reconstruction remains above the random-latent control range (11–15 dB in the companion supplementary study). The pilot

TABLE III
REAL IBM HARDWARE RECONSTRUCTION PILOT (BACKEND `ibm_fez`, 256 SHOTS/BASIS, NO RETRAINING). PSNR COMPARES HARDWARE-DECODED AND SIMULATOR-DECODED RECONSTRUCTIONS AGAINST THE SAME GROUND-TRUTH IMAGE.

| Image | Patches | Sim. PSNR (dB) | HW PSNR (dB) |
|---|---|---|---|
| Monalisa (HVK1D) | 16 | 37.99 | 25.90 |
| CIFAR cat (HVK2D) | 16 | 44.85 | 31.52 |
| CIFAR ship, hydrofoil (HVK2D) | 16 | 36.56 | 26.44 |
| CIFAR ship, sea boat (HVK2D) | 16 | 42.57 | 31.20 |
| CIFAR airplane (HVK2D) | 16 | 46.79 | 29.10 |
| CIFAR mean $\pm$ std (4 images) | – | $42.69 \pm 3.84$ | $29.56 \pm 2.03$ |

TABLE IV
REAL-HARDWARE ANCHOR POINTS: SAME CHECKPOINTS, NO RETRAINING, ON TWO FURTHER BACKENDS. THE ORIGINAL PILOT'S `ibm_fez` POINT IS SHOWN FOR REFERENCE.

| Image | Backend | Shots | PSNR (dB) |
|---|---|---|---|
| Monalisa | ibm_fez (original pilot) | 256 | 25.90 |
| Monalisa | ibm_marrakesh | 256 | 25.94 |
| Monalisa | ibm_kingston | 1024 | 26.10 |
| CIFAR cat | ibm_fez (original pilot) | 256 | 31.52 |
| CIFAR cat | ibm_kingston | 256 | 28.93 |
| CIFAR cat | ibm_kingston | 1024 | 31.24 |

used 176 circuits; the contemporaneous account-meter reading increased by 16 quantum-processor seconds, with the provenance limitation documented in the supplement.

Five images constitute a pilot, and we make no hardware-advantage claim. The narrower result is that the trained HVK observable channel remains visually and quantitatively informative in these device executions under the reported circuit and shot budget. Full methodology — exact gate sequences, per-basis measurement construction, job IDs, and pre-submission local-validation results — is documented in the companion supplementary document.

The companion Supplementary Material reports a further hardware-validation exercise built on this same checkpoint-replay methodology: fresh HVK1D checkpoints trained at $N \in \{4, 6, 8\}$, replayed epoch-by-epoch on real IBM Quantum hardware and on the IonQ cloud simulator [26], in support of the training-dynamics diagnostic discussed in Section V-A and the Supplementary Material. We keep it out of the main text because it is a single-patch, single-seed probe in service of a diagnostic whose own negative control (below) does not establish sensitivity to the Hamiltonian term it is meant to track.

### *E. Noise/Shot Trade-off: A Free Simulator Complement*

The five-image pilot above used a single shot budget (256/basis) and one execution per image, which bounds how much it can say about noise sensitivity on its own. Real IBM Quantum time is a scarce free-tier resource, so we separate what a *calibrated device-noise simulation* can establish at zero hardware cost from what genuinely requires hardware. Using `qiskit-aer` with the live calibration snapshot of `ibm_fez` (`FakeFez`, no quantum-processor time consumed), we replay the same already-trained checkpoints' circuits at four shot budgets (256/512/1024/4096 shots per basis, three repeated executions each) and compare against both the noiseless-simulator ceiling and the real-hardware point already reported above.

Two findings follow, averaged over the five checkpoints. First, shot noise accounts for only part of the noiseless-to-real gap: moving from the noiseless ceiling to a calibrated-noise simulation at the same 256-shot budget already costs 8.68 dB on average, before any real device is involved. Second, more shots does not reliably recover it: increasing to 4096 shots per basis changes PSNR by only $-0.05$ dB on average across the five checkpoints, with individual images ranging from a 1.33 dB gain to a 0.96 dB loss — at this circuit size and shot range, shot count is not a lever that reliably improves reconstruction quality. Comparing the 256-shot noisy simulation directly against the real-hardware point isolates what the calibration snapshot does *not* capture: a further 4.24 dB average gap (range 2.25–8.06 dB across images), attributable to device effects a single calibration snapshot does not fully capture — drift between calibration and execution, crosstalk, and correlated errors chief among them. The airplane image is the clear outlier (8.06 dB residual gap, roughly double any other image), which we report rather than exclude; a single-execution, five-image pilot cannot distinguish an image-specific effect from ordinary job-to-job variation. This calibrated-noise simulation is a free, repeatable complement to the pilot, not a substitute for it: it bounds how much of the observed hardware gap is attributable to modeled device noise versus unmodeled effects, using zero additional quantum-processor time.

### *F. Real-Hardware Anchor Points: Repeated Execution and a Second Backend*

The simulated trade-off above is bounded, in turn, by whether it actually reflects real devices. We spent a small, explicitly quota-tracked slice of the free-tier IBM Quantum allowance (four jobs, well under 1% of the monthly limit) confirming it directly: the same two checkpoints (*Monalisa*, CIFAR *cat*) replayed with no retraining on `ibm_marrakesh` and `ibm_kingston` — two backends distinct from the original pilot's `ibm_fez` — at the original 256-shot budget and, separately, at 1024 shots.

*Monalisa* reproduces almost exactly across backends at matched shots (25.94 vs. 25.90 dB, `ibm_marrakesh` vs. `ibm_fez`), and gains only 0.16 dB from quadrupling shots to 1024 — consistent with the simulated finding above that shot count is not a reliable lever. The CIFAR *cat* point is the more informative one: on `ibm_kingston` at the original 256-shot budget it reaches 28.93 dB, 2.59 dB below the `ibm_fez` pilot point, while its own 1024-shot execution on the same backend recovers to 31.24 dB, within 0.28 dB of the original. This is real job-to-job and backend-to-backend variation of a magnitude comparable to the unmodeled residual identified by the simulator comparison (Section IV-E) — direct evidence, not merely a simulated bound, that a single 256-shot execution on a single backend is not yet a stable estimate of this pipeline's hardware performance, and that repeated


Fig. 2. *Monalisa* reconstructed from real IBM Quantum hardware measurements (right) against the same trained checkpoint's noiseless-simulator replay (center) and ground truth (left). The hardware image is visibly noisier and exhibits patch-boundary banding, while the subject remains recognizable.

TABLE V
$D_4$ EQUIVARIANCE ERROR FOR UNPOOLED AND POOLED OBSERVABLE-GRID MAPS (LOWER IS BETTER; POOLED MAP IS GROUP-AVERAGED OVER ALL EIGHT SQUARE SYMMETRIES).

| Model | Mean error | Std. error |
|---|---|---|
| $D_4$-pooled HVK2D | $9.57 \times 10^{-17}$ | $1.26 \times 10^{-17}$ |
| No positional encoding | $7.40 \times 10^{-1}$ | $1.52 \times 10^{-1}$ |
| HVK2D positional | $8.15 \times 10^{-1}$ | $1.29 \times 10^{-1}$ |
| Local/raw | $8.37 \times 10^{-1}$ | $1.31 \times 10^{-1}$ |

execution matters at least as much as backend choice or shot count individually.

### G. Exact $D_4$-Equivariant Observable Pooling

A $D_4$-pooled HVK2D observable-grid map is equivariant to the eight symmetries of the square *by construction*: $\Phi_{D_4}(x) = \frac{1}{8}\sum_{g \in D_4} P_g^{-1}\Phi(gx)$ averages the unpooled map over the group orbit, so $\Phi_{D_4}(hx) = P_h \Phi_{D_4}(x)$ for every $h \in D_4$ up to floating-point error. We verify this numerically over 7000 patch transforms of cached CIFAR images (Table V, Figure 5): the pooled map's equivariance error is $9.57 \times 10^{-17} \pm 1.26 \times 10^{-17}$ (floating-point noise), while the original unpooled positional map is not measurably more symmetric than a local/raw control ($0.815 \pm 0.129$ vs. $0.837 \pm 0.131$) — confirming the unpooled map is only $D_4$-*motivated* by the patch grid's geometry, not empirically equivariant, and that only the explicit pooling construction delivers the property. This is currently a feature-grid-level construction and numerical diagnostic rather than a component of the trainable reconstruction pipeline; integrating it end-to-end (Section V-F) is concrete future work.

### H. Entanglement Necessity in the Nonlocal-Structure Protocol

This diagnostic is deliberately constructed to reward pair-observable structure: the target is a fixed synthetic function of *simulated* patch features not concatenated into any candidate's own input block (the companion supplement gives the leakage-audit procedure that rules out circularity), and every control receives the same raw inputs as HVK2D. Table VI and Figure 6 report the result across five seeds: HVK2D's entangling observable channel reaches mean $R^2 = 0.9735$ (32.77 dB PSNR), a 17.74 dB improvement over a no-entanglement control ($R^2 = 0.019$) and 18.52 dB over random-VQC latents ($R^2 = -0.173$); freezing the quantum circuit (features fixed, only readout trained) still reaches $R^2 = 0.853$, while a parameter-matched classical control ($R^2 = -0.030$) and a frozen-classical control ($R^2 = -0.903$) fail outright. This is evidence that an entangling pair-observable channel is *representationally necessary within the tested feature class and fixed linear-readout protocol* to fit this target.

### I. A Scoped Training-Dynamics Diagnostic (Full Account in the Supplementary Material)

Beyond the results above, we tracked two exploratory training-dynamics readouts: a magnetization-style statistic $M_z(t)$ and a signed energy-to-entanglement ratio $R_{ES}(t)$. Neither exhibits a sharp transition — both drift smoothly over training with run-to-run fluctuations, and any apparent onset is a within-run median-plus-$2\sigma$ threshold crossing rather than a calibrated statistical test. The negative control settles the reading: the identical protocol applied to a classical-replacement variant, with the Hamiltonian energy identically zero throughout training, crosses that threshold in all $4/4$ tested cases — exactly as often as the Hamiltonian-regularized model's $4/4$ — so we make no transition claim and report these traces solely as an interpretability readout on the learned energy and entanglement (full protocol and tables in the companion Supplementary Material).

## V. Discussion

### A. Why the Physics-Informed Regularizer Was Inert

Table VII shows the Hamiltonian energy term is, if anything, a mild drag on reconstruction quality in the present system. These are fresh, same-environment measurements (*Monalisa*, 200 steps, seed 42); an earlier draft reported different absolute PSNR values for these same three configurations, produced

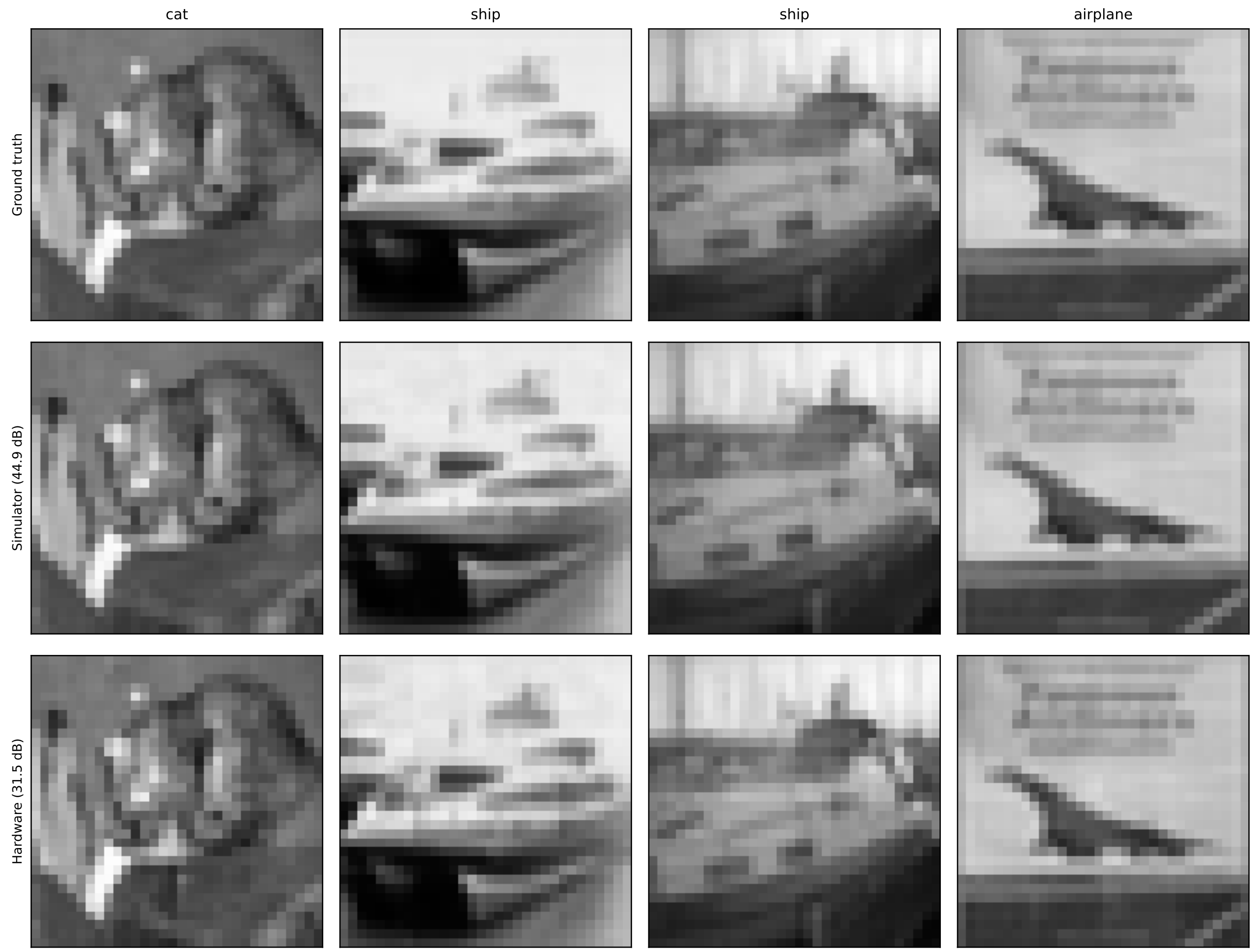


Fig. 3. Four CIFAR-10 images reconstructed from real IBM Quantum hardware measurements (bottom row) against noiseless-simulator replays of the same trained HVK2D checkpoints (middle row) and ground truth (top row).

TABLE VI
HVK2D RESTRICTED PAIR-CORRELATION DIAGNOSTIC ACROSS FIVE SEEDS. DELIBERATELY FAVORABLE TO PAIR-OBSERVABLE STRUCTURE; EVIDENCE FOR ENTANGLEMENT-NECESSITY UNDER A MATCHED LINEAR READOUT.

| Model | Mean MSE | Mean PSNR (dB) | Mean $R^2$ |
|---|---|---|---|
| HVK2D entangling observables | $8.5969 \times 10^{-4}$ | 32.77 | 0.9735 |
| Freeze quantum | $4.7904 \times 10^{-3}$ | 27.34 | 0.8533 |
| No entanglement | $3.1461 \times 10^{-2}$ | 15.03 | 0.0191 |
| Raw-linear classical | $3.1654 \times 10^{-2}$ | 15.00 | 0.0133 |
| Parameter-matched classical | $3.3033 \times 10^{-2}$ | 14.82 | $-0.0297$ |
| Random VQC | $3.7616 \times 10^{-2}$ | 14.25 | $-0.1732$ |
| Freeze classical | $6.0973 \times 10^{-2}$ | 12.17 | $-0.9025$ |

by a script that was never committed to the repository and cannot be recovered or audited, so those earlier numbers are superseded here rather than cited (full account of the discrepancy, plus additional observable-sector controls and follow-up experiments, in the Supplementary Material). The legacy objective adds a signed mean energy to the reconstruction loss; because learned energy can go negative, the optimizer can lower total loss by driving energy down without improving reconstruction, and removing the term improves PSNR from $38.63$ to $44.56$ dB. A contrastive variant — assigning lower energy to correct feature–position pairs than to shuffled pairs — improves over the signed-energy baseline ($42.92$ dB) but remains below its own no-energy-loss counterpart ($44.56$ dB). We read this as a case where the reconstruction loss already dominates the optimization signal at this model scale and dataset size: a physically motivated regularizer has room to help when it resolves an otherwise underdetermined objective, and here the pixel-fidelity loss alone is sufficiently informative that the extra term mostly adds noise to the gradient. This does not mean Hamiltonian regularization is never useful — only that its value is conditional on the reconstruction loss being under-constrained, a condition this patch-level pixel

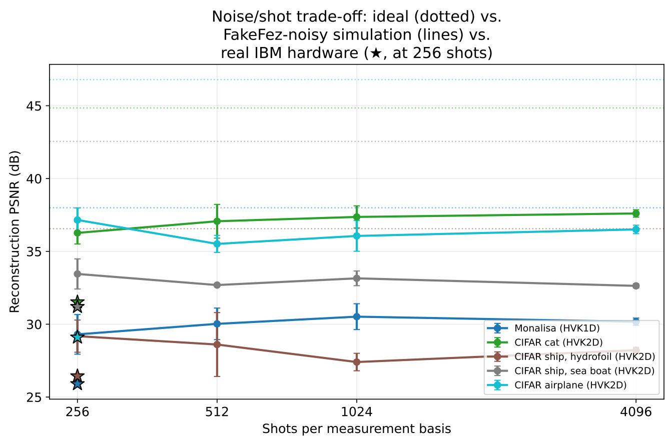


Fig. 4. Calibrated device-noise simulation (`FakeFez`, mean $\pm$ std over three executions). Dotted lines show noiseless ceilings and stars show real-hardware measurements. Increasing 256 to 4096 shots changes mean PSNR by only $-0.05$ dB.

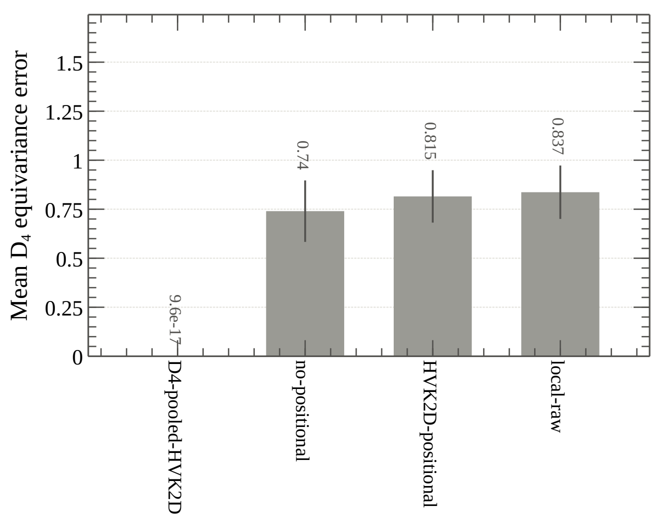


Fig. 5. $D_4$ equivariance diagnostic. The unpooled positional map is not equivariant; the group-pooled map is equivariant to numerical precision.

task does not meet. The Supplementary Material additionally reports a root-cause fix (bounding the couplings and using a positive/squared energy loss, which stops the term from hurting but does not make it help) and an exploratory decoder-feature variant that improves 2 of 3 seeds but regresses sharply on the third, too seed-sensitive to report as a validated result here.

The training-dynamics readouts of Section IV-I provide a descriptive window into how the learned energy and entanglement evolve, and nothing more: they are not calibrated statistical tests and must not be read as thermodynamic transitions.

### *B. Task-Dependent Representational Scope*

Whether a quantum feature map or kernel offers any advantage over a classical model is a property of the specific data and target, not a fixed property of the model class, and dequantization results have repeatedly shown that apparent quantum advantages on structured, low-entanglement classical data can be matched by classical algorithms once the right classical feature is identified [27], [44]; where provable separations do exist, they are established for carefully constructed learning problems rather than for generic structured data [45]. The restricted diagnostic of Section IV-H is constructed so that the relevant structure is a genuine distant-element product raw or local features cannot represent under a fixed linear readout, and that is precisely the regime in which HVK's entangling channel separates. Natural image patches, at the resolution and patch size used in ordinary reconstruction, are a different regime — low-entanglement, locally structured data where a linear or shallow classical readout is already close to sufficient. The companion supplementary study's held-out comparisons (Section IV-C) characterize exactly this second regime; together, the results delineate where the tested entangling representation does and does not yield a measurable benefit.

### *C. HVK as a Representative Instance of a Broader Design Pattern*

The boundary reported in Section IV-C raises an immediate question: is it a property of HVK's particular implementation choices, or of the broader class of architectures HVK instantiates? HVK combines three ingredients that recur, individually, throughout the hybrid quantum-vision literature rather than being idiosyncratic to this codebase: matrix-product-state compression of structured input into a classical or hybrid feature vector [7], [9], [11], [22]; a Pauli-correlator (measured-observable) latent space handed to a classical decoder rather than a raw quantum state, in the spirit of quantum-enhanced feature maps [21]; and a physically motivated auxiliary regularizer alongside a separate task loss [19]. HVK is not a minimal or contrived combination of these three choices assembled to produce a particular result; each is independently well-motivated and widely used, and HVK's specific instantiation (six-qubit VQC, 19–27-D Pauli-correlator vector, graph-Hamiltonian energy term) sits well within the parameter ranges typical of this literature rather than at an unusual extreme.

This representativeness matters for ruling out an uncharitable alternative reading of the held-out result (Section IV-C) and the inert-regularizer result (Section V-A): that HVK's specific implementation choices — an unmotivated qubit count, an idiosyncratic latent width, or an unusual regularizer formulation — rather than the design pattern itself, are responsible for tying classical baselines, and that a different, non-representative choice within the same pattern would have won. Because HVK's instantiation sits well within the ordinary parameter ranges of this literature rather than at an unusual extreme, that alternative reading is not supported: nothing about HVK's specific engineering choices explains away the result. Per the data-dependent framing of Section V-B [27], whether any instance of this design pattern outperforms a resource-matched classical alternative is a property of the specific data and target, not of the pattern's architectural motivation, so we do not claim the measured boundary transfers unchanged to every dataset,

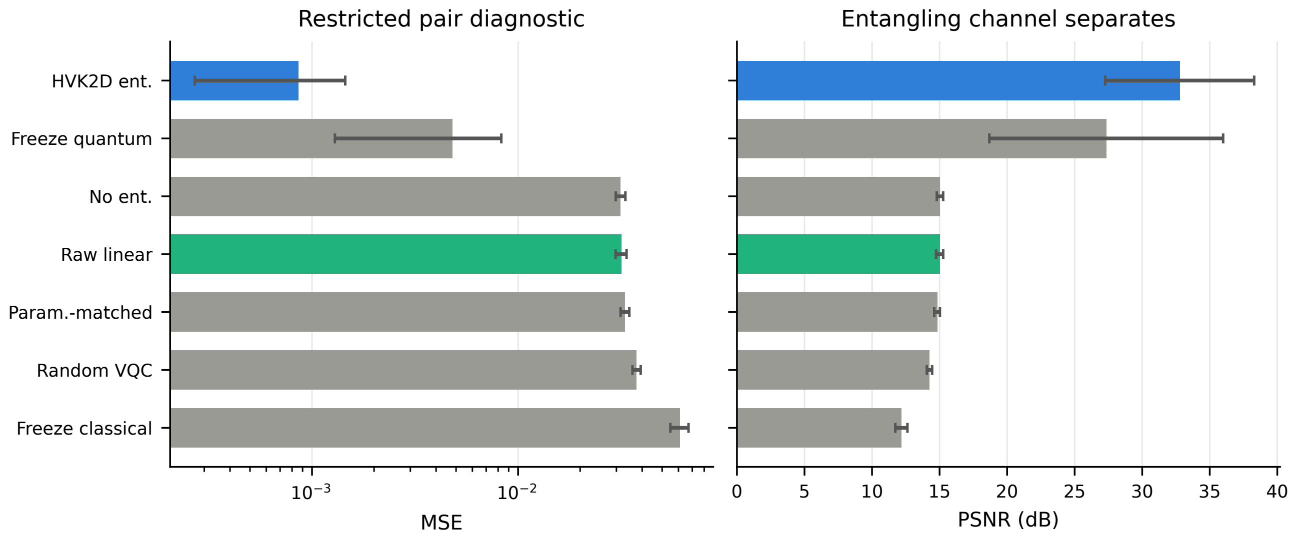


Fig. 6. Restricted pair-correlation diagnostic summary (mean $\pm$ std over five seeds). The entangling observable feature map dominates every control on this deliberately favorable task.

TABLE VII
HAMILTONIAN-SECTOR CONTROLS (SINGLE-SEED, FRESH SAME-ENVIRONMENT REPRODUCTION: *Monalisa*, 200 STEPS, SEED 42). THE ENERGY OBJECTIVE IS DIAGNOSTICALLY USEFUL BUT DOES NOT IMPROVE RECONSTRUCTION ACCURACY. SEE THE SUPPLEMENTARY MATERIAL FOR THE FULL REPRODUCIBILITY INVESTIGATION AND ADDITIONAL OBSERVABLE-SECTOR CONTROLS.

| Experiment | PSNR (dB) | Interpretation |
|---|---|---|
| Legacy baseline, signed energy | 38.63 | Standard energy-regularized run |
| No energy loss | 44.56 | Removing energy improves PSNR |
| Contrastive Hamiltonian core | 42.92 | Stronger objective helps modestly vs. baseline, still below no-energy-loss |

patch size, or qubit count. What this section establishes is narrower and, we think, more useful: the architecture class HVK instantiates — MPS feature compression, a Pauli-correlator latent space, and a physics-informed regularizer — is sound and general, a fair and properly-scaled test of the pattern rather than a strawman version of it, so its measured behavior here is evidence worth taking at face value rather than dismissing as an idiosyncrasy of one implementation.

### D. Topology Alignment as an Inductive Bias

Aligning the latent interaction graph with image-patch adjacency (HVK2D vs. HVK1D) is a plausible architectural inductive bias. The companion supplementary study tests this hypothesis directly under matched held-out conditions. Its real-circuit subset (3 seeds, 3 held-out CIFAR-10 images) gives an HVK1D-minus-HVK2D mean PSNR difference of $0.16$ dB; after averaging images within seed, the bootstrap 95% CI is $[-0.38, 0.46]$ dB and the seed-level Wilcoxon test gives $p = 0.50$. HVK2D completes the matched 90-step runs in roughly half the wall-clock time. A broader five-seed surrogate comparison across CIFAR-10, Fashion-MNIST, and PathMNIST likewise finds numerically small topology differences (at most $0.07$ dB in mean PSNR). At the tested scale, topology therefore appears to be primarily a resource-cost choice rather than a reliable accuracy advantage; these tests do not establish formal equivalence.

### E. Architectural Differentiation Matrix

Table VIII contrasts HVK's structural properties against reference frameworks, including the three nearest hybrid-vision architecture families discussed above (register-based QAE, quantum-kernel vision, TN-circuit classifier). Its distinguishing features — a Pauli-correlator latent space, 2D Fourier positional bias with enforced $D_4$ pooling, and a Hamiltonian energy diagnostic — are documented and scoped in the preceding sections.

### F. Validated Scope and Next Steps

- Every reconstruction result in this paper (Sections IV-A–IV-B) is per-image trained, not held-out generalization; the companion supplementary study's held-out protocol is the appropriate evidence for a generalization claim, and its finding that local/raw maps match or exceed the tested HVK map under the shared readout should be read alongside the architecture-level results here.
- The $D_4$-pooled map (Section IV-G) is currently a feature-grid-level construction and numerical diagnostic, not yet integrated into the trainable end-to-end reconstruction pipeline; doing so is concrete future work (Section VI).
- The entanglement-necessity result (Section IV-H) is scoped to a task deliberately constructed to require nonlocal structure; it does not by itself establish an advantage on ordinary natural-image reconstruction (see Section V-B).

TABLE VIII
ARCHITECTURAL DIFFERENTIATION FROM REPRESENTATIVE BASELINE DESIGNS, INCLUDING THE THREE NEAREST HYBRID-VISION ARCHITECTURE FAMILIES (REGISTER-BASED QAE, QUANTUM-KERNEL VISION, TN-CIRCUIT CLASSIFIER).

| Axis | Conv. AE baseline | GAN baseline | Register-based QAE | Quantum-kernel vision | TN-circuit classifier | HVK (ours) |
|---|---|---|---|---|---|---|
| Latent representation | Continuous $\mathbb{R}^d$ | Continuous $\mathbb{R}^d$ | Quantum register | Scalar kernel value | MPS/circuit state | Pauli-correlator vector |
| Spatial bias | Convolutional | Convolutional | Encoding-dependent | Encoding-dependent | MPS chain ordering | 2D Fourier encoding |
| Auxiliary objective | Bottleneck/reconstruction | Adversarial loss | Fidelity/reconstruction | None (kernel value only) | Classification loss | Hamiltonian diagnostic |
| Enforced group symmetry | Not in baseline | Not in baseline | Not in reference design | Not in reference design | Not in reference design | $D_4$ observable pooling |
| Evaluated role | Reconstruction | Reconstruction | Compression | Classification | Classification | Reconstruction + diagnostics |

- The hardware pilot (Section IV-D) is a five-image, single-trial, unmitigated pilot, not a statistically resolved hardware benchmark.
- The training-dynamics traces (Section IV-I) are an interpretability readout, not evidence of a transition: the threshold's negative control fires on a classical-replacement variant with the Hamiltonian energy identically zero exactly as often as on the Hamiltonian-regularized model (4/4 vs. 4/4, full table in the Supplementary Material), so the readout has no demonstrated sensitivity to the Hamiltonian term itself.
- The matched real-circuit topology comparison uses only 3 seeds and 3 held-out images per topology; the broader five-seed comparison is surrogate-based. These results support a resource-cost interpretation but do not establish formal equivalence of the topologies.
- Every result here is at a six-qubit, depth-18 scale. We therefore do not extrapolate these findings to wider circuits: the trainability of variational models is known to degrade with width and depth through barren-plateau effects [46], [47], so whether the tested boundary between HVK and resource-matched classical maps moves at larger scale is an open question this study does not answer.

## VI. CONCLUSION

We presented the Hamiltonian Vision Kernel, a new hybrid quantum–classical architecture for image reconstruction that assembles matrix-product-state patch features, a measured Pauli-correlator latent space, enforced $D_4$-equivariant observable pooling, and a learnable Hamiltonian energy diagnostic into a single pipeline — a combination not previously assembled in this form (Section V-C). Across six real image datasets, per-image-trained HVK2D models reconstruct with high fidelity, and on a resource-matched, multi-seed held-out comparison a pre-declared TOST equivalence test shows HVK2D is *competitive with*, rather than inferior or superior to, simple classical controls (Section IV-C): local/raw maps match or exceed it under the same fitted readout, and TOST confirms the gap is small enough to call the two practically tied. We further reported a real IBM Quantum hardware reconstruction pilot: replaying trained checkpoints without retraining yielded $25.90$–$31.52$ dB PSNR from hardware measurements, establishing feasibility at a five-image, single-trial scope rather than a general hardware-performance claim.

If HVK ties classical baselines on ordinary reconstruction, its case rests on the capabilities a purely classical reconstruction pipeline does not provide: a restricted entanglement-sensitive result on a task requiring nonlocal structure, where pair observables are necessary within the tested fixed-readout feature class ($R^2 = 0.9735$ vs. $R^2 \leq 0.02$ for non-entangling controls), an interpretable Hamiltonian energy diagnostic, and native execution on real quantum hardware. Alongside these, the assembled pipeline is exactly $D_4$-equivariant by construction (equivariance error $9.57 \times 10^{-17}$); we report that as a design-correctness property rather than a quantum capability, since the same Reynolds averaging applied to a classical feature map is equally exact. None of these is in tension with the tie-with-classical result: entangling observables separate only on the tested nonlocal target, and group pooling fixes equivariance without by itself establishing an accuracy gain. We make no claim of quantum advantage. HVK is, at this scope, a new architecture that works, is competitive with resource-matched classical baselines, runs on real quantum hardware, and is equipped with an entanglement-sensitive channel, an interpretable Hamiltonian diagnostic, and native hardware execution — capabilities a classical reconstruction pipeline does not natively expose. We consider this carefully measured boundary a constructive contribution alongside the architecture itself.

Future work should integrate the $D_4$-pooled map into an end-to-end trainable pipeline (currently a feature-grid-level construction, Section V-F); scale the hardware pilot toward a statistically resolved benchmark with error mitigation and repeated trials; search for further real-world tasks with genuine nonlocal structure where HVK's restricted entanglement-sensitive separation translates into a reconstruction advantage; and extend the resource-matched, leakage-audited evaluation protocol used in the companion supplementary study to other hybrid quantum vision architectures.

## DATA AND CODE AVAILABILITY

The source code, experiment drivers, machine-readable result summaries, validation arrays, and figure inputs supporting this study are included in the accompanying repository. The file `project_artifacts/submission_claim_audit.json` maps each numerical headline claim to its retained source artifact — including the held-out comparison the tested HVK map loses and the withdrawn training-dynamics material — and automated tests verify those mappings. Dataset provenance and access routes are cited in the manuscript and detailed in the supplementary artifact map. IBM Quantum job identifiers and reconstructed outputs are retained; as disclosed in the supplement, the raw account-usage service response was not serialized, so the reported quota change is an execution-log record rather than a repository-recomputable measurement.

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